ANALYZING MACHINE LEARNING MODELS FOR PREDICTING POWER CONSUMPTION: CASE OF STUDY IN TETUAN CITY, MOROCCO

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Abstract – Accurate energy forecasting is vital for managing rising global energy demands and climate change. This study evaluates the performance of three machine learning models—XGBoost, Random Forest, and Long Short-Term Memory (LSTM) networks—in predicting short-term energy consumption in Tetuan City, Morocco. Using a dataset from the UCI Machine Learning Repository, the research examines the impact of time-based features on prediction accuracy. XGBoost, with the lowest Root Mean Squared Error (RMSE), outperforms Random Forest and LSTM, particularly after hyperparameter tuning. The results highlight the significance of model selection and feature engineering, with XGBoost proving most robust for energy consumption forecasting.

Key words: energy forecasting, machine learning, XGBoost, Random Forest, LSTM, Tetuan City.

I. INTRODUCTION

As global energy demands continue to rise and the effects of climate change become more pronounced, the need for accurate and efficient energy forecasting has never been more critical. The energy sector accounts for approximately three-quarters of global greenhouse gas emissions, highlighting its pivotal role in addressing climate change [1].

Achieving net-zero emissions by 2050 is essential to limit the rise in global temperatures to 1.5°C, necessitating a profound transformation in how energy is produced, transported, and consumed. This transformation involves a significant shift towards renewable energy sources, improvements in energy efficiency, and the widespread adoption of new technologies.

Accurate energy forecasting aids in planning and implementing these changes by predicting future energy needs and helping to manage resources more effectively. The Global Energy Perspective 2023 report underscores the importance of such forecasting, projecting diverse energy transition scenarios that influence global temperature increases and energy demands across various sectors [2].

Consequently, numerous predictive models have been developed to forecast energy consumption, leveraging advanced machine learning and statistical techniques to enhance accuracy and reliability.

A. Energy Forecasting

Energy demand forecasting can be broadly categorized into short-term, medium-term, and long-term forecasting. Short-term forecasting typically spans from a few minutes to a few days and is crucial for operational decisions such as load scheduling and real-time energy management. Techniques used for short-term forecasting often include machine learning models like Support Vector Regression (SVR), Artificial Neural Networks (ANNs), and recurrent neural networks such as Long Short-Term Memory (LSTM) networks [3].

Medium-term forecasting, covering weeks to months, aids in maintenance scheduling, budgeting, and resource allocation. Models used for this type of forecasting may include seasonal ARIMA models, hybrid approaches like CNN-LSTM, and ensemble methods that combine multiple forecasting techniques to enhance robustness and accuracy [4].

Long-term forecasting, which extends over several years, is essential for strategic planning, policymaking, and investment decisions. Long-term models often rely on econometric and physical simulation methods, integrating factors such as economic growth, technological advancements, and climate policies to predict future energy demands [5].

In addition to these categories, forecasting methods can also be classified based on their approach: physical-based models and data-driven models. Physical-based models use mathematical representations of physical processes and are particularly effective for long-term predictions and scenarios where detailed system behaviour needs to be modelled. Conversely, data-driven models rely on historical data and statistical techniques to predict future trends. These models include time series analysis, machine learning, and deep learning methods, which have gained popularity due to their ability to handle large datasets and uncover complex patterns [6].

While Support Vector Regression and Artificial Neural Networks are widely used due to their strong predictive performance in various scenarios, this study focuses on evaluating XGBoost, Random Forest, and Long Short-Term Memory (LSTM) networks. These models were selected for their proven effectiveness in handling diverse data types and addressing specific forecasting challenges. XGBoost is known for its robustness and efficiency in dealing with structured data and capturing nonlinear relationships. Random Forest is valued for its ability to reduce overfitting through ensemble learning and its interpretability. LSTM networks are particularly suitable for sequential data and time series forecasting because of their capability to capture temporal dependencies and patterns.

Given the critical role of accurate energy forecasting in managing energy consumption and planning for future needs, this study aims to explore and evaluate these machine learning models to identify the most effective approach for short-term energy consumption prediction in Tetuan City, Morocco.

B. Research Objective and Questions

The primary objective of this work is to develop and evaluate predictive models for forecasting energy consumption in Tetuan City, enhancing the accuracy and reliability of predictions to support effective energy management and planning. In this study we focus on answering two main questions:

- How do time-based features impact the accuracy of energy consumption forecasts in Tetuan City?
- Which machine learning model (XGBoost, Random Forest, or LSTM) provides the most accurate predictions for short-term energy consumption in Tetuan City?

By leveraging machine learning techniques, we aim to enhance the accuracy and reliability of energy demand forecasts, which are crucial for effective energy management and planning in the face of rising global energy demands and climate change.

II. METHODOLOGY

In this section, we outline the methodology used for data management and model development to forecast energy consumption in Tetuan City. The process is structured into three main parts: data description, data processing, and model training and evaluation.

A. Data description

The dataset used in this study is sourced from the UCI Machine Learning Repository [5]. This dataset is related to the power consumption of three different distribution networks in Tetuán City, located in northern Morocco. It is a multivariate, time-series dataset with applications in social science and is typically used for regression tasks.

The dataset contains power consumption data recorded at regular intervals. The primary features include temperature, humidity, Wind Speed, General Diffuse Flows, Diffuse Flows and the power consumption of 3 zones (Zone 1, Zone 2 and Zone 3) (Fig. 1).

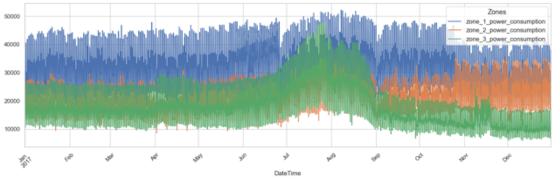

Fig. 1 Energy Consumption (MWh) for Zone 1,2 and 3

B. Data Processing

In the data processing phase, several essential steps were undertaken to prepare the dataset for accurate energy consumption forecasting. The date column was converted to a datetime format, and the dataset columns were standardized. Missing values were addressed through mean imputation, preserving the dataset's integrity without introducing significant biases. Furthermore, normalization using Min-Max scaling was applied to temperature, humidity, wind speed, and diffuse flow features, ensuring uniformity across features and facilitating the convergence of machine learning algorithms.

C. Feature Engineering

To enhance the predictive power of the dataset, various time-based and statistical features were created, Time-based features such as hour, day of the week, month, and year were added, this helps in understanding the cyclical nature of energy consumption. Lag features and rolling statistics for power consumption were created for the power consumption of each zone. These engineered features are crucial for capturing temporal patterns and dependencies within the data, thereby enhancing the predictive capabilities of the models.

D. Exploratory Data Analysis

- Energy Distribution

The energy consumption distributions across the three zones exhibit a right skewed pattern, with Zone 1 showing the highest and most variable consumption, peaking between 30,000 and 35,000 MWh. Zone 2's consumption is less variable and peaks between 20,000 and 25,000 MWh, while Zone 3 shows a broader range, peaking between 15,000 and 20,000 MWh (Fig. 2) (Fig. 3). These patterns suggest that Zone 1 has more sporadic and higher energy usage, potentially due to larger industrial or commercial activities, whereas Zone 2's more consistent consumption might indicate residential or smaller-scale commercial use. The variability and peaks in Zone 3 indicate mixed usage patterns.

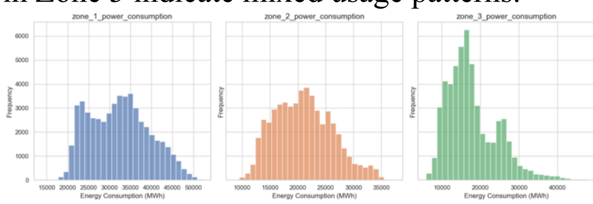

Fig. 2 Histogram Energy consumption per zone

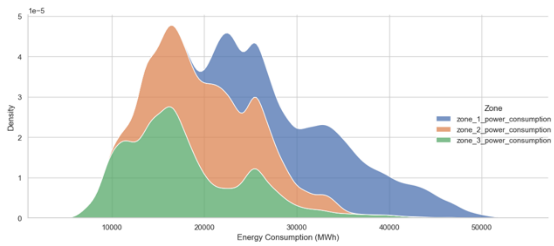

Fig. 3 Density of Energy Consumption Distribution per Zone

- *Temporal patterns*

The analysis of energy consumption patterns revealed distinct trends on both an hourly and monthly basis. The hourly trend demonstrates two prominent consumption peaks, occurring between 12:00-16:00 hours and 18:00-22:00 hours, reflecting higher energy demands during midday and evening periods (Fig. 4). For the other side, the monthly trend highlights a clear seasonality, with a significant increase in energy consumption during the summer months of July and August (Fig.5). This surge can be attributed to higher cooling demands due to elevated temperatures, leading to increased use of air conditioning systems. Understanding these temporal patterns is crucial for optimizing energy supply strategies and improving the accuracy of consumption forecasts.

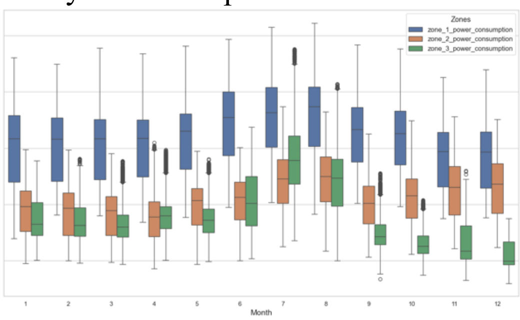

Fig. 4 Energy Consumption by hour

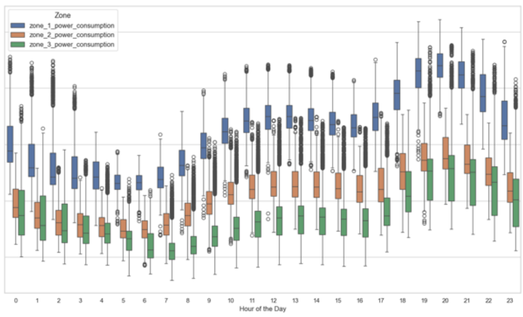

Fig. 5 Energy Consumption by month

- *Time series decomposition*

The time series decomposition for energy consumption in three zones of Tetuan City reveals detailed insights into the underlying patterns and structures in the data. Each time series is decomposed into its observed, trend, seasonal, and residual components. This decomposition shows a pronounced peak in the trend component around midyear, indicating a common seasonal influence on energy demand, likely attributable to higher temperatures and the consequent increased use of cooling systems (Fig. 6).

Each zone demonstrates distinct daily cyclic patterns in the seasonal component, reflecting consistent and predictable daily energy usage behaviors. However, Zone 3 exhibits a notably higher amplitude in its seasonal component, suggesting a greater variability in daily energy consumption.

This indicates that Zone 3 experiences more significant fluctuations in daily energy usage, potentially due to more dynamic or varied activities within this zone. Additionally, the residual component for Zone 3 shows the highest degree of variability among the three zones, pointing to more frequent or impactful irregular events. This higher residual variability suggests that Zone 3 is subject to more unplanned activities or is more sensitive to external factors compared to Zones 1 and 2.

The identification and understanding of these components are critical for the development of precise and reliable energy consumption forecasting models.

- Correlation

The correlation matrix for energy consumption across the three zones in Tetuan City reveals that Zone 1 and Zone 2 exhibit a strong positive correlation (0.83), indicating that their energy usage patterns are closely related, likely due to similar influencing factors such as industrial or commercial activities. Zone 1 and Zone 3 show a moderately strong correlation (0.75), suggesting a significant relationship but with distinct factors also affecting Zone 3's energy consumption. The weakest correlation is observed between Zone 2 and Zone 3 (0.57), indicating that Zone 3's energy consumption patterns are somewhat independent of Zone 2, potentially due to a mix of residential and commercial activities in Zone 3 compared to more homogeneous activities in Zone 2.

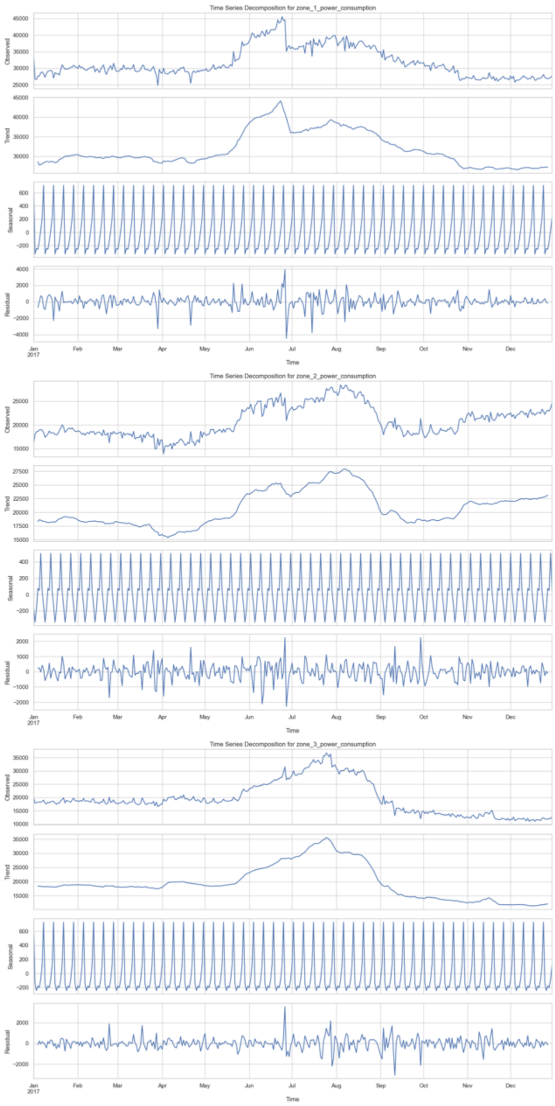

Fig. 6 Time Series Descomposition

E. Model Development

For the modelling in this project, an appropriate data splitting strategy that respects the temporal nature of the dataset was crucial. Unfortunately, only one year of data is available, which limits the ability to fully capture seasonality and long-term trends. Despite this limitation, a strategy was employed to split the dataset into training and test sets using a 70/30 ratio (Fig. 7). This method ensures a substantial portion of the data is reserved for training while maintaining an independent test set for evaluating the model's performance on unseen data. The training set comprises the earlier time periods, and the test set includes the later periods.

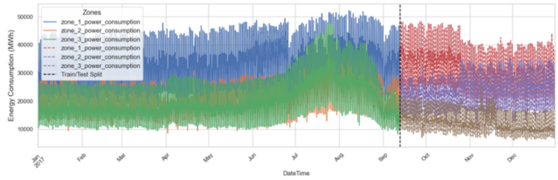

Fig. 7 Data Split for Modeling

To develop the forecasting models, multiple modelling techniques were explored, including XGBoost, Random Forest, and Long Short-Term Memory (LSTM) networks.

1) XGBOOST

This model was selected for its high performance and efficiency in handling structured data, making it well-suited for energy consumption forecasting. Its ability to build robust models using gradient boosting techniques makes it a strong candidate.

- *Model configuration: Defining the Baseline Model and Cross-Validation*

The process began with configuring a baseline XGBoost model with 1000 estimators and a learning rate of 0.002.

Cross-validation was conducted using Timeseries Split with 5 splits to maintain the temporal order of the data. The negative mean squared error (neg_mean_squared_error) was used as the scoring metric. The baseline model achieved a cross-validated RMSE of approximately 1731.86 MWh.

- *Model training and Initial Evaluation*

The baseline model was trained on the entire training dataset. Early stopping was set to 50 rounds to prevent the model from overfitting by monitoring performance on a validation set.

Predictions were then made on the test set, and the RMSE for each target variable was calculated:

- Zone 1: 1301.09 MWh
- Zone 2: 775.18 MWh
- Zone 3: 1990.80 MWh

To improve the model's performance, hyperparameter tuning was performed using RandomizedSearchCV. The parameters explored included:

- n_estimators: [100, 500, 1000]
- learning_rate: [0.01, 0.1, 0.2]
- max_depth: [3, 5, 7]
- subsample: [0.8, 1.0]
- colsample_bytree: [0.8, 1.0]

The best hyperparameters identified were:

{'subsample': 0.8, 'n_estimators': 500, 'max_depth': 5, 'learning_rate': 0.1, 'colsample_bytree': 1.0}

These hyperparameters were selected based on their ability to minimize the cross-validated RMSE.

- *Evaluation of the Tuned Model*

The tuned model was then trained using the entire training dataset with the identified best hyperparameters.

Cross-validation was again conducted using TimeSeriesSplit, and the tuned model achieved a cross-validated RMSE of approximately 667.56 MWh.

The tuned model's performance on the test set was evaluated, resulting in the following RMSE values for each zone:

- Zone 1: 384.43
- Zone 2: 250.65
- Zone 3: 752.83

Significant improvements in RMSE were observed for all three zones with the tuned model. The improvements are as follows: Zone 1 saw an RMSE reduction from 1301.09 to 384.43, an improvement of 916.66. Zone 2 experienced a decrease in RMSE from 775.18 to 250.65, resulting in an improvement of 524.53. Zone 3 also showed a notable improvement, with the RMSE decreasing from 1990.80 to 752.83, an enhancement of 1237.97. These results indicate that the tuned model significantly improved the accuracy of energy consumption forecasts across all zones.

Zone	*Baseline Model*	*Tuned Model*	*Improve*
Zone 1	*1301.09*	*384.43*	*916.66*
Zone 2	*775.18*	*250.65*	*524.53*
Zone 3	*1990.80*	*752.83*	*1237.97*

Table 1 XGBoost Model Evaluation

2) RANDOM FOREST

Random Forest was chosen for its robustness in handling high-dimensional data and its ensemble approach, which mitigates overfitting. This model's ability to aggregate the predictions of multiple decision trees makes it suitable for the complex task of energy consumption forecasting.

- *Model configuration*

The baseline model, set with default parameters, was trained on the entire training dataset. Upon evaluation, the RMSE for each zone was calculated:

- Zone 1: 758.43 MWh
- Zone 2: 3021.85 MWh
- Zone 3: 1533.64 MWh

To enhance the model's performance, a comprehensive hyperparameter tuning process was undertaken using RandomizedSearchCV. The hyperparameters explored included:

- n_estimators: [100, 300, 500]
- max_depth: [10, 20, 30]
- min_samples_split: [2, 5, 10]
- min_samples_leaf: [1, 2, 4]
- max_features: ['sqrt', 'log2']

The model underwent numerous iterations, each tested through cross-validation using TimeSeriesSplit with 3 splits, ensuring temporal integrity. After extensive search, the best parameters identified were

{'n_estimators': 300, 'max_depth': 20, 'min_samples_split': 5, 'min_samples_leaf': 2, 'max_features': 'sqrt'}

With the best parameters identified, a finer hyperparameter tuning was performed using GridSearchCV. This process further honed the model by testing slight variations around the identified optimal parameters:

- n_estimators: [290, 300, 310]
- max_depth: [18, 20, 22]
- min_samples_split: [4, 5, 6]
- min_samples_leaf: [1, 2, 3]
- max_features: ['sqrt']

The fine-tuning process yielded the following best parameters:

{'n_estimators': 300, 'max_depth': 20, 'min_samples_split': 5, 'min_samples_leaf': 2, 'max_features': 'sqrt'}

- *Model training and Evaluation*

The tuned Random Forest model was then trained on the entire training dataset. Cross-validation using TimeSeriesSplit was conducted, resulting in a cross-validated RMSE of approximately 1168.63 MWh.

Upon evaluation on the test set, the tuned model achieved the following RMSE values:

- Zone 1: 911.48 MWh
- Zone 2: 2610.25 MWh
- Zone 3: 1887.58 MWh

A detailed comparison between the baseline and tuned models revealed the following improvements:

Zone	*Baseline Model*	*Tuned Model*	*Improve*
Zone 1	*758.43*	*911.48*	*-153.06*
Zone 2	*3021.85*	*2610.25*	*411.60*
Zone 3	*1533.64*	*1887.58*	*-353.94*

Table 2 Random Forrest Model Evaluation

3) LSTM Networks

This is a type of recurrent neural network (RNN) specifically designed to capture long-term dependencies in sequential data. It is particularly well-suited for time series forecasting as it can learn from the temporal dynamics and patterns in the data, making it highly effective for capturing complex relationships over time.

- *Model configuration*

The configuration of the LSTM model began with data normalization. To ensure consistent feature scaling, the MinMaxScaler was applied to both the input features and the target variables. This normalization process was crucial for the LSTM network, which is sensitive to the scale of the input data.

Next, the normalized data was transformed into sequences to capture temporal dependencies. Each sequence consisted of 20-time steps, a choice aimed at providing the model with sufficient context from historical data to make accurate predictions.

The LSTM model architecture included:

- LSTM Layers: Two LSTM layers were configured, with the first layer consisting of 128 units and the second layer consisting of 64 units. Both layers employed the ReLU activation function and L2 regularization to mitigate overfitting. The first LSTM layer was set to return sequences to allow the second LSTM layer to process the entire sequence.

- Batch Normalization: Applied after the LSTM layers to stabilize and accelerate the training process.

- Dropout Layer: A dropout layer with a rate of 0.3 was included to prevent overfitting by randomly setting a fraction of input units to 0 at each update during training.

- Dense Layer: The output layer was a dense layer with three units, corresponding to the three target variables (energy consumption in Zones 1, 2, and 3).

The model was compiled using the Adam optimizer with a learning rate of 0.0005 and mean squared error (MSE) was used as the loss function.

- *Model training and Evaluation*

The LSTM model was trained using strategies like early stopping and learning rate reduction to optimize performance and prevent overfitting:

- Early Stopping: This monitored the validation loss and halted training if no improvement was observed for 10 consecutive epochs, ensuring that the model did not overfit to the training data.

- Learning Rate Reduction: If the validation loss plateaued for 5 epochs, the learning rate was reduced by a factor of 0.5, with a minimum learning rate set to 0.0001.

The training involved a maximum of 50 epochs with a batch size of 64, allowing the model to learn effectively. The following results were obtained upon evaluating the model's performance using Root Mean Squared Error (RMSE):

- Zone 1**:** 864.41 MWh
- Zone 2**:** 1117.61 MWh
- Zone 3**:** 958.32 MWh

These results indicate that the LSTM model performed reasonably well across all zones, with Zone 2 exhibiting the highest RMSE, suggesting it had the most complex consumption patterns to capture. However, the improvements in RMSE highlight the model's effectiveness in learning from the temporal data and making accurate predictions.

III. DISCUSSION AND RESULTS

In this study, the performance of three machine learning models—XGBoost, Random Forest, and LSTM—was evaluated for forecasting energy consumption across three zones in Tetuán City.

The XGBoost model demonstrated notable performance in forecasting energy consumption. After hyperparameter tuning, the model showed a significant improvement in RMSE, reducing the cross-validated RMSE from 1731.86 to 667.56.

Zone-specific RMSEs also showed improvement, with Zone 1 and Zone 2 exhibiting reduced errors. The tuned model achieved the following RMSE values for each zone:

- Zone 1: 384.43 MWh
- Zone 2: 250.65 MWh
- Zone 3: 752.83 MWh

Significant improvements were observed in all zones:

- Zone 1: RMSE reduced from 1301.09 to 384.43, an improvement of 916.66 MWh.
- Zone 2: RMSE reduced from 775.18 to 250.65, an improvement of 524.53 MWh.
- Zone 3: RMSE reduced from 1990.80 to 752.83, an improvement of 1237.97 MWh.

The comparative analysis of RMSE values across different zones suggests that while XGBoost effectively captures consumption patterns for all zones, it achieved the most substantial improvements for Zone 3.

The Random Forest model, despite being a robust ensemble method, did not perform as well as XGBoost. The tuned model configuration resulted in RMSE values of:

- Zone 1: 911.48 MWh
- Zone 2: 2610.25 MWh
- Zone 3: 1887.58 MWh

While the tuned model showed improvement in Zone 2, it performed less optimally for Zones 1 and 3, indicating challenges in capturing the variability and complexity of energy consumption patterns in these zones. The improvements were as follows:

- Zone 1: RMSE increased from 758.43 to 911.48, a decline of -153.06 MWh.
- Zone 2: RMSE decreased from 3021.85 to 2610.25, an improvement of 411.60 MWh.
- Zone 3: RMSE increased from 1533.64 to 1887.58, a decline of -353.94 MWh.

Finally, the LSTM model, designed to handle sequential data, performed reasonably well but faced challenges with the dataset. The RMSE values for each zone were:

- Zone 1: 864.41 MWh
- Zone 2: 1117.61 MWh
- Zone 3: 958.32 MWh

These results indicate that the LSTM model performed adequately across all zones, with Zone 2 exhibiting the highest RMSE, suggesting it had the most complex consumption patterns to capture. Despite attempts to optimize the model architecture and training process, the LSTM struggled to capture the underlying patterns effectively.

In comparative terms, XGBoost emerged as the most effective model for this task, demonstrating the lowest RMSE values across most zones. Random Forest followed, performing adequately in less variable scenarios but lagging in more complex patterns. The LSTM, although a strong candidate for time-series data, did not perform to the same level, highlighting the importance of model selection based on data characteristics.

IV. RESULTS

The models were able to capture the general trends in power consumption, with XGBoost showing the highest accuracy. The models were assessed based on their Root Mean Squared Error (RMSE) to determine their accuracy and effectiveness.

The following sections summarize the performance of each model.

Subsequently, a graph is presented with the results comparing actual and predicted power consumption using the XGBoost model, indicating robust performance in capturing general energy usage trends across all zones.

- In Zone 1, the model shows strong alignment between actual and predicted values, successfully capturing peaks and troughs.

Zone	Model	Baseline RMSE	Tuned RMSE
Zone 1	XGBoost	1301.09	384.43
	Random Forest	758.43	911.48
	LSTM		864.41
Zone 2	XGBoost	775.18	250.65
	Random Forest	3021.85	2610.25
	LSTM		1117.61
Zone 3	XGBoost	1990.80	752.83
	Random Forest	1533.64	1887.58
	LSTM		958.32

Table 3. Model performance per zone

- Zone 2 also demonstrates high accuracy, with predictions closely following actual consumption patterns, reflecting substantial RMSE improvement.

- Although Zone 3 shows notable RMSE improvement, the predictions exhibit more variability compared to actual values, suggesting a need for further refinement to fully capture its complex consumption dynamics.

Overall, the XGBoost model stands out as the most effective for energy consumption forecasting in Tetuán City, demonstrating the lowest RMSE values and consistently capturing consumption patterns across different zones. This evaluation underscores the importance of selecting appropriate machine learning models tailored to specific data characteristics and forecasting needs for accurate and reliable predictions in energy management and planning.

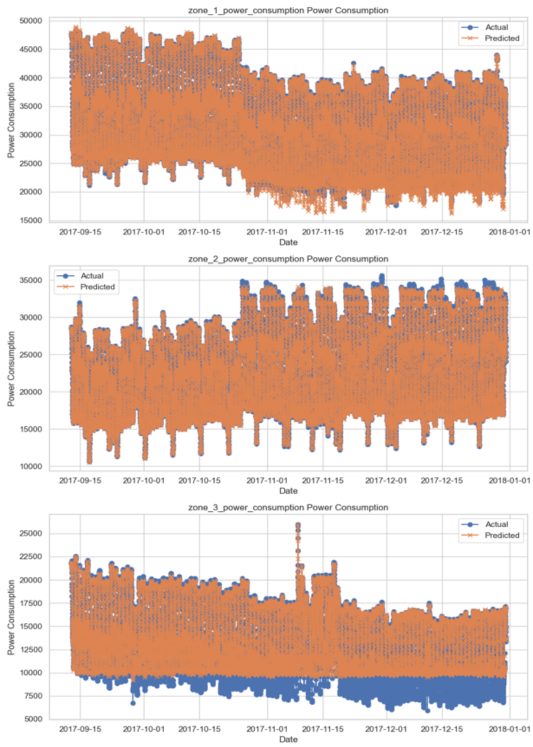

Fig. 6 XGBooster Real Data vs. Predicted Data

V. CONCLUSION

The primary objective of this study was to develop and evaluate predictive models for forecasting energy consumption in Tetuan City to enhance the accuracy and reliability of these predictions and support effective energy management and planning. The research addressed two main questions: the impact of time-based features on forecast accuracy and the comparative performance of XGBoost, Random Forest, and LSTM models.

- *Key findings:*

Impact of Time-Based Features: Time-based features, including seasonal and daily cycles, significantly influenced the accuracy of energy consumption forecasts. Models that effectively incorporated these temporal patterns demonstrated superior predictive performance.

- *Model Performance:*

- **XGBoost:** This model emerged as the most accurate for short-term energy consumption forecasting in Tetuan City, achieving the lowest RMSE across all three zones after hyperparameter tuning. XGBoost's robustness in handling structured data and leveraging gradient boosting techniques contributed to its superior performance.
- **Random Forest:** While showing reasonable predictive capabilities, Random Forest had higher RMSE values compared to XGBoost, particularly in Zone 3. Despite some performance improvements from hyperparameter tuning, the model struggled to capture complex consumption patterns as effectively as XGBoost.
- **LSTM:** Although well-suited for capturing temporal dependencies, the LSTM model did not perform as well as the tree-based models. The higher RMSE values suggested difficulties in optimizing the model for the specific characteristics of this dataset and forecasting task

Implications:

- The use of machine learning techniques, particularly XGBoost, significantly enhances the accuracy and reliability of energy demand forecasts. This is crucial for effective energy management and planning, especially in the context of rising global energy demands and climate change.
- XGBoost's ability to consistently capture consumption patterns across different zones underscores its potential as a robust tool for energy consumption forecasting.

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